Venkata Lakshmi Parimala Pasupuleti

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github.com/venkatalakshmiparimala/AgentProbe | Remote - US

PROFESSIONAL SUMMARY

AI Engineer with 9 years of production Python engineering, specialized in **Claude-powered agentic systems**, RAG evaluation harnesses, and enterprise AI governance. Built AgentProbe using **Anthropic Claude API + LangGraph** -- implementing RAG pipelines, prompt/context strategies, compliance guardrails, and trace-level observability. Delivered 100% PHI-compliant AWS data pipelines and HIPAA-compliant AI on clinical records -- demonstrates regulated-environment AI deployment from day one. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

WORK EXPERIENCE

George Mason University
Aug 2024 -- Aug 2025

Research Associate -- Virginia, USA

- Built regression models ($R^2 = 0.977$) in Python and R across 190+ countries; applied Isolation Forest to automate data-quality checks across 3 research pipelines -- surfaced anomalies manual review missed.
- Designed Tableau, ggplot2, and Quarto dashboards translating ML outputs into faculty decisions -- 40% comprehension improvement across two cohorts.

Entrata
Jan 2017 -- Aug 2024

Software Engineer

- Pioneered Entrata's AI engineering foundation -- built and internally shipped the first AI pipeline (Python, PyTorch, OpenAI API) across **1.6M real financial records** with output validation, quality gates, and compliance controls; later productized by the team into a customer-facing feature.
- Reduced deployment times 40% building 20+ microservices (Python, PostgreSQL, RabbitMQ, FastAPI) for **100K+ monthly active users** across 120+ Agile sprints; led Docker containerization cutting execution time 50%; improved query performance 30% across 200+ SQL queries.
- Achieved **100% PHI compliance** migrating all client data to AWS S3 with Libsodium-encrypted pipelines -- zero security incidents; built affordable housing compliance modules (Section 8/HCV, LIHTC, HUD) - income eligibility verification, AMI threshold calculations, and HUD regulatory reporting across 100K+ unit portfolios; eliminated 80% of manual workload via Python/PL-SQL automation.

PROJECTS

AgentProbe: Claude-Powered Agentic AI Toolkit
Open Source

Python · LangGraph · LangChain · Anthropic (Claude) · FastAPI · ChromaDB · OpenAI · Groq · React · Docker · GitHub Actions -- github.com/venkatalakshmiparimala/AgentProbe

- Built with **Anthropic Claude API + LangGraph**: 3-agent runtime (Orchestrator, RAG Tracer, Auto Fix) with tool calling, inter-agent state handoffs, and multi-provider model routing -- deployed as 15-endpoint FastAPI backend with CI/CD quality gates.
- Implemented **RAG evaluation harnesses + compliance guardrails**: 5-metric LLM-as-judge (factual accuracy, hallucination, retrieval, completeness, compliance) blocks non-compliant runs before production; React dashboard tracks per-run retrieval quality, latency, and regression drift.

AI on FHIR: NLP-Driven Medical Records Query System

Flask · LLMs · FHIR · React · spaCy · BERT · Embeddings · Docker -- github.com/venkatalakshmiparimala/AI_on_FHIR

- Reduced clinical retrieval time ~60% by building an NLP platform translating natural language queries (EN/ES) to FHIR APIs -- **HIPAA-compliant RBAC + AI governance** across structured healthcare records; enterprise AI in a regulated environment from day one.

PropLens: Lease Intelligence & Compliance AI
In Progress

Python · LangGraph · ChromaDB · FastAPI · LLMs (Anthropic/OpenAI) · React

- Building Claude-powered RAG pipeline over lease PDFs -- LangGraph agents, LLM-as-judge scoring, and compliance gap detection across 1,000+ tenant records; reference architecture for enterprise agentic AI in a regulated domain.

EDUCATION

M.S. Data Analytics Engineering -- George Mason University — Graduated May 2026

GPA: 3.87 · Fairfax, Virginia

B.Tech Computer Science & Engineering -- Jawaharlal Nehru Technological University, Anantapur — 2013 -- 2017

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, R, Bash, HTML/CSS

AI & Agents: Anthropic API (Claude), LangGraph, LangChain, RAG Pipelines, LLM Evaluation, LLM-as-Judge, Evaluation Harnesses, Agent Observability, Prompt Engineering, AI Governance, Multi-Agent Orchestration, ChromaDB, Embeddings, OpenAI API, Groq, Hugging Face, FastAPI

ML & Data: PyTorch, TensorFlow, Scikit-learn, XGBoost, Isolation Forest, BERT, spaCy, NumPy, Pandas, Tableau, ggplot2

Cloud & Infrastructure: AWS (S3, EC2, RDS, Glue), Azure, GCP, Docker, CI/CD (GitHub Actions), RabbitMQ, PostgreSQL, MySQL